

LOKESH VARMA KONDURU

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SUMMARY

Software Engineer with 3+ years building distributed systems, event-driven microservices, and cloud-native applications across fraud detection, Agri-tech, and e-commerce using **Java/Spring Boot**, **Node.js/Express**, and **React/TypeScript**. Deployed production platforms serving **1,000+ users** on **Kafka**, **Kubernetes**, and **AWS**, with strong expertise in **Python/FastAPI** and AI-powered agentic systems using **MCP tool calling**, **RAG pipelines**, and **LangChain**. M.S. Computer Science, **Virginia Tech** (GPA: **4.0/4.0**).

EDUCATION

Virginia Tech — M.S. in Computer Science

GPA: 4.0 / 4.0 | Aug 2024 – May 2026

Mahindra University — B.Tech. in Computer Science

GPA: 8.28 / 10 | Jul 2019 – May 2023

PROFESSIONAL EXPERIENCE

Software Engineer

Virginia Tech

Sep 2024 – May 2026

- Cut **bug turnaround time by 40%** by building a full-stack platform with **React/TypeScript**, Next.js SSR, and PostgreSQL, establishing **Jest** unit and integration test standards validated on every PR through **GitHub Actions** CI/CD.
- Reduced deployment time **from ~30 minutes to under 10 minutes** by containerizing services with **Docker and Kubernetes** on **AWS EC2**, eliminating manual steps with Terraform IaC, automated rollback policies, and continuous deployment on every merge.
- Shipped **5+ Google Earth Engine geospatial tools** for satellite imagery processing, NDVI visualization, and soil-vegetation-climate analysis, used by **1,000+ researchers and growers** across Virginia Tech Extension with Prometheus/Grafana observability. [Website](#)
- Boosted geospatial data pipeline throughput by **40%** by building an async **Python/FastAPI** backend with **Redis** caching and PostgreSQL, improving performance through query optimization, connection pooling, and efficient async data access patterns.
- Built an **AI-powered agentic advisory assistant** by integrating LangChain, LangGraph, and FAISS into **Java/Spring Boot** REST APIs, implementing **RAG pipelines** and **MCP tool calling** to deliver real-time contextual agroclimate recommendations.

Software Engineer

Marine Corps Community Services (MCCS)

Sep 2024 – Dec 2025

- Scaled fraud detection to handle **10K+ transactions** by architecting microservices with **Node.js/Express**, **MongoDB**, and **Redis**, designing data models, service contracts, and caching layers for real-time fraud score and transaction lookups. [\[GitHub\]](#)
- Reduced fraud signal processing latency from **~600ms to ~360ms** by orchestrating **Kafka** event streaming on **Kubernetes**, optimizing through consumer group tuning, partition scaling, and **OpenTelemetry**-guided bottleneck resolution with CloudWatch tracing.
- Reduced manual fraud review time by **30%** by building a **GenAI investigation assistant** with LangChain, LangGraph, and FAISS, implementing **RAG pipelines** and **MCP tool calling** to automate intelligent fraud signal analysis and decision support.

Software Engineer

DuoDecimal Tech LLP

Jul 2023 – Jul 2024

- Delivered **4 production platforms** across events, bookings, hostel, and food management using React/TypeScript, **Node.js/Express**, and MongoDB with JWT/OAuth2 authentication, mentoring junior engineers on **code quality**, **testing standards**, and **PR review**.
- Improved booking efficiency by **30%** and cut response latency **from ~400ms to under 200ms** by building a real-time ordering platform with **Kafka** async event streaming, integrating **Easebuzz** and **Razorpay** payment processing with Redis-backed session caching.
- Reduced manual paperwork by **60%** for **25,000+ users** by developing an RFID-integrated hostel management system with Node.js/Express, building **MongoDB** indexed queries, event-driven access control, and automated Jest testing on every release.
- Enabled zero-downtime releases for **25,000+ users**, reducing release cycles by **65%** through staged rollouts, health checks, and rollback policies on **AWS EC2** behind ALB with automated **GitHub Actions** CI/CD pipelines.

Software Engineer Intern

MindGraph Technologies

Jan 2023 – Jun 2023

- Improved data visibility by **40%** for Telecom and CPG clients by building analytics dashboards with **Python/Flask** and Plotly Dash, integrating **Azure AD SSO** for secure enterprise access and automated deployments with Azure DevOps CI/CD on Microsoft Azure.
- Ensured data quality and reliability across **10M+ airline records** by designing distributed pipelines with **PySpark**, **Kafka**, and Apache Airflow, reducing pipeline failure rate by **40%** through automated PyTest unit and integration tests and built-in alerting.

PROJECTS

Nova Clinic | Python/FastAPI, MySQL, HTML5/CSS, JWT

Feb 2026 – May 2026

- Built **JWT-authenticated REST APIs** over HTTPS with **Python/FastAPI** and **MySQL**, implementing secure endpoints, input validation, and role-based middleware for a responsive HTML5/CSS appointment management frontend. [\[GitHub\]](#) | [Live](#)
- Designed a normalized **MySQL** schema with **role-based access control** enforcing data privacy through JWT claim-based permissions, covering all critical endpoints with PyTest integration tests and validating security through structured access control testing.

E-Commerce Bookstore Platform | Java/Spring Boot, Spring Security, MySQL, Redis, React

Feb 2025 – May 2025

- Delivered catalog, cart, and checkout workflows using **Java/Spring Boot**, Spring Security, and MySQL with JWT/OAuth2 authentication, achieving **sub-200ms API response times** through Redis caching, connection pooling, and optimized JPA query design. [\[GitHub\]](#) | [Live](#)
- Improved recommendation accuracy for a **1,000+ book** catalog by integrating a **LangChain**-powered assistant with **FAISS** semantic search and **Redis** session caching, reducing search latency through vector index optimization and prompt engineering for user flows.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C, C++

Frontend: React.js, Next.js SSR, Redux Toolkit, React Query, Tailwind CSS, Socket.io, Plotly Dash

Backend: Java/Spring Boot, Spring Security, Node.js/Express, FastAPI, Flask, REST APIs, Microservices, Distributed Systems, JWT/OAuth2

Databases & Caching: PostgreSQL, MongoDB, MySQL, Redis, Firebase

AI Integration: LangChain, LangGraph, OpenAI API, RAG Pipelines, FAISS, MCP Tool Calling

Cloud & DevOps: AWS (EC2, S3, API Gateway, ALB, CloudWatch), Kubernetes, Docker, Kafka, PySpark, Apache Airflow, Terraform, GitHub Actions CI/CD, Microsoft Azure, Azure DevOps, OpenTelemetry, Prometheus/Grafana, Linux

Testing & Tools: Jest, PyTest, JUnit, Selenium, Postman, TDD, Maven/Gradle, Git, Jira, Google Earth Engine, Swagger